

Individual Contribution — Cedric Fernolend

Cedric Fernolend — Individual Contribution Statement

Primarily responsible for: repository setup and Git workflow, dataset research and evaluation, sourcing and integrating the COVID-19 data, the reproducible `code/` pipeline, the project proposal, the final report structure, and publication via GitHub Pages.

Repository setup and tooling

created the repository and connected it to RStudio, establishing the project structure and version-control workflow we used throughout ([d4fcc75](#), [a55e718](#), [8ee3454](#)). Later in the project I introduced a branch-and-pull-request workflow

Dataset research and evaluation

set up the dataset-research documents and investigated several candidate datasets, notes into Quarto files, reorganising the file tree ([e754d3e](#), [05faed6](#), [956311f](#), [5854271](#)).

US Supply Chain candidate initial analysis and documentation ([3204012](#), [85e50e6](#), [71d8c88](#)). ([f562a4c](#)).

Initial data analysis of the mental-health dataset

carried out the exploratory analysis of the Household Pulse mental-health data ([08a1b3e](#), [9daf4f4](#), [44c595e](#), [df51671](#)).

Project proposal

wrote and iterated on the project proposal ([3b015b5](#), [fd850d7](#), [a501297](#), [06de582](#)). When the proposal was updated, I narrowed our scope to two focused research questions with named outputs and documented the risks of each candidate dataset.

Sourcing and integrating the COVID-19 data

When we committed to the mental-health direction, the project needed a second dataset that did not yet exist in the repository. I sourced state-level COVID-19 case data from the New York Times, verified its licence, and built the linkage to the survey data ([a8d5e17](#), [1917614](#)). This required several non-obvious steps: differencing the NYT *cumulative* counts into daily new cases, normalising by 2020 Census population to make states comparable (cases per 100,000), and joining each state's daily cases into the correct survey window using a **range join** on the window start and end dates

Repository restructure and reproducible pipeline

I migrated the project to the required course template layout and built the reproducible pipeline ([51a4ea9](#))

Reproducibility with renv

committed the lockfile so package versions are pinned across machines ([d1fb4d3](#), PR #10), including verifying that only the lockfile and activation scripts are tracked while the package library itself stays out of version control. This was a shared task, because we had issues with git. So I ended up uploading it.

Report structure and limitations

I restructured the final report so that each research question presents only the question, method and outputs, with all interpretation moved into a dedicated section, and wrote the limitations ([70c03a1](#)). identifying the Omicron windows as high-leverage observations, and documenting the suppressed estimates as **not missing at random**. For RQ2 I added correlation metrics alongside the plots and showed that the apparent co-movement.

Publication and presentation

I published the project as a Quarto website via GitHub Pages. ([9b69040](#), [3db7dc0](#)).

Reflection

The most useful lesson for me was that the hardest problems in this project were rarely statistical. Sourcing a second dataset that could actually be *joined* to the first, matching state names, aligning daily case counts to two-week survey windows, and normalising by population so large states did not dominate took considerably more thought than fitting the regression that followed.

I also learned to distrust a result that looks too convenient. The supply-chain dataset appeared to have a strong risk predictor until we checked and found it was derived from the outcome itself. Later, the level correlation in RQ2 looked convincing until differencing showed it was largely a shared trend.

Finally, restructuring the repository mid-project taught me the value of a good workflow, using branches, pull requests and issues.